Controlled State-Dimensionality Experiment for Weekly Threshold Control in Credit Scoring

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1 Introduction

This document summarizes the current repository state after the controlled state-dimensionality update of the weekly threshold-control reinforcement learning framework for credit scoring. The simulator remains weekly, synthetic, and portfolio-oriented: the controller does not approve individual applications directly, but instead updates weekly score thresholds for new and repeat applicants. The accepted cohort then produces delayed repayments, defaults, and possible late recoveries, so the quality of a policy can only be judged on the full portfolio trajectory rather than on immediate cohort profit.

The present experiment asks one narrow question: how does controller behavior change when the observation size is increased from 12 to 20, 30, and 50 features? The design is intentionally controlled. Environment dynamics, delayed outcomes, delayed rewards, warm-up logic, terminal settlement, action semantics, seeds, scenarios, training profile, evaluation protocol, bootstrap settings, controller set, and baseline set are held fixed. Only `state_dim` changes.

2 Controlled Experimental Design

2.1 Fixed Components

The following items are identical across all four runs:

- weekly simulator dynamics and scenario set;
- new/repeat split-policy logic;
- threshold range from 35 to 85 with step 5;
- reward formulation with delayed reward and constraint penalties;
- warm-up phase, interactive horizon, and terminal settlement;
- seeds 11, 23, and 37 in the active quick profile;
- controller set: `dqn`, `double_dqn`, `ppo`, `sac`, `static_threshold`, `profit_oriented`, `risk_aware_weekly`, `constraint_aware_weekly`, and `split_policy_static`;
- bootstrap confidence interval logic and exported metrics.

2.2 Varying Component

The only factor that changes is the observation dimensionality:

$$\texttt{state_dim} \in \{12, 20, 30, 50\}.$$

The first 12 ordered features are identical in every configuration. Higher-dimensional states append deterministic features derived only from already observed simulator history, last-week portfolio statistics, rolling windows, or lagged threshold behavior. No future information is introduced.

3 Observation Layers

The state space is expanded cumulatively.

State size	Composition
12D	Unchanged baseline: week progress, overall and segment approval rates, rolling realized default rate, expected current default rate, realized profit, rolling profit volatility, projected capital usage, outstanding ratio, and normalized new/repeat thresholds.
20D	Baseline 12D plus 8 additional features: repeat-share, expected profit and expected NPV per application, realized NPV, weekly reward, threshold gap, capital headroom, and realized-vs-expected default gap.
30D	20D plus 10 historical features: lagged approval/default/profit/capital usage, rolling means and standard deviation for recent profit, and lagged threshold changes for new and repeat segments.
50D	30D plus 20 segment and stability features: segment-specific approval/default/profit indicators, accepted-share ratios, reward and capital rolling volatility, cumulative reward/profit to date, outstanding-capital changes, threshold-gap history, and current application-volume ratio.

Table 1: Cumulative state construction. Exact ordered definitions are stored in `state_dimension_manifest.md`.

The implementation verifies programmatically that the first 12 coordinates are unchanged by comparing the 12D baseline against 20D, 30D, and 50D under a common reset and fixed action sequence.

4 Experimental Protocol

The exported repository outputs correspond to the `quick` profile. The key settings are:

- 8 warm-up weeks and 26 interactive weeks;
- 260 applications per week in the quick profile;
- 12 training episodes per RL seed;
- 4 evaluation runs per seed and scenario;
- 95% bootstrap confidence intervals with 120 resamples;
- six evaluation scenarios: `base_market`, `drift`, `adverse_stress`, `class_imbalance_shift`, `noise`, and `split_policy_dynamics`.

Each dimension is written to its own directory:

- `outputs/dim_12/`
- `outputs/dim_20/`

- `outputs/dim_30/`
- `outputs/dim_50/`

Cross-dimension summary files are written to the root `outputs/` directory.

5 Results

5.1 Best Overall Controller by Dimension

Table 2 shows the strongest controller when baselines and RL agents are pooled together.

State dim	Best overall	Type	Cumulative reward
12	`profit_oriented`	baseline	88 080.60
20	`profit_oriented`	baseline	88 080.60
30	`profit_oriented`	baseline	88 080.60
50	`profit_oriented`	baseline	88 080.60

Table 2: Best overall controller by dimension from `outputs/overall_best_by_dimension.csv`.

The result is itself informative. Because the non-RL baselines do not depend on the observation vector, the best overall baseline remains identical across dimensions. This confirms that the experiment isolates the effect of state size on the RL controllers rather than silently changing the comparison set.

5.2 Best RL Controller by Dimension

State dim	Best RL	Expected profit	NPV	Cumulative reward	Default rate
12	`dqn`	59 197.47	57 144.61	67 308.36	0.1233
20	`double_dqn`	61 096.44	58 978.98	69 812.02	0.1260
30	`dqn`	66 758.29	64 755.07	78 219.69	0.1185
50	`dqn`	66 946.81	64 815.05	78 198.39	0.1195

Table 3: Best RL controller by dimension from `outputs/best_rl_by_dimension.csv`.

Three patterns follow from Table 3.

- Increasing the state from 12D to 20D improves the best RL profit and reward modestly.
- The largest step occurs between 20D and 30D, where the best RL controller improves both profitability and default rate.
- The move from 30D to 50D adds almost no extra reward: expected profit increases only slightly, cumulative reward is effectively flat, and default rate ticks up. The gains therefore saturate rather than continue linearly.

5.3 Cross-Dimension Figures

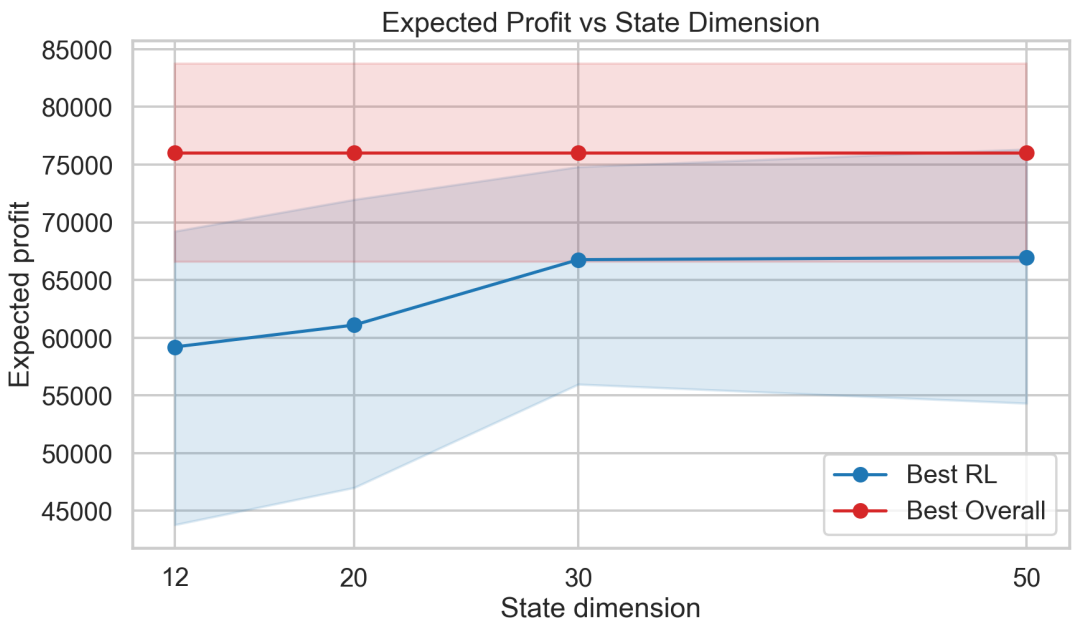


Figure 1: Expected profit versus state dimension for the best RL controller and the best overall controller.

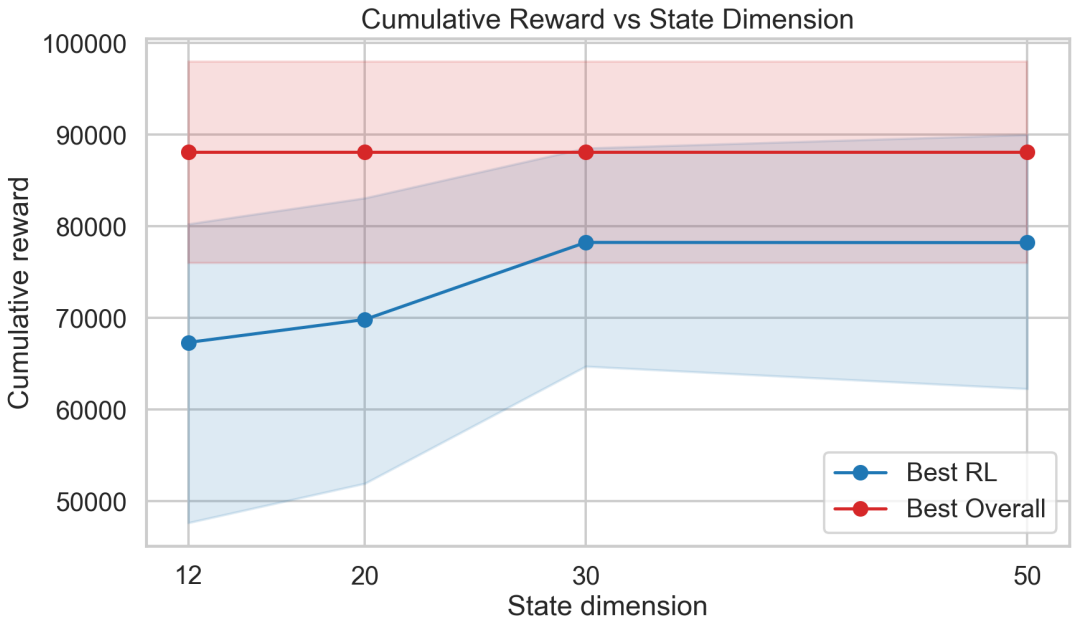


Figure 2: Cumulative reward versus state dimension.

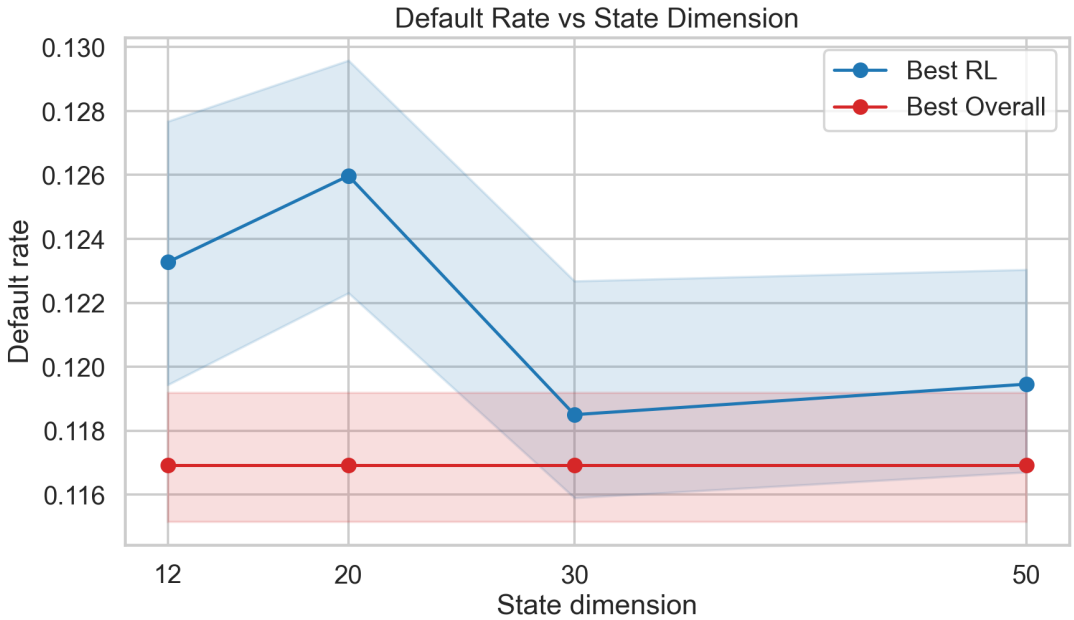
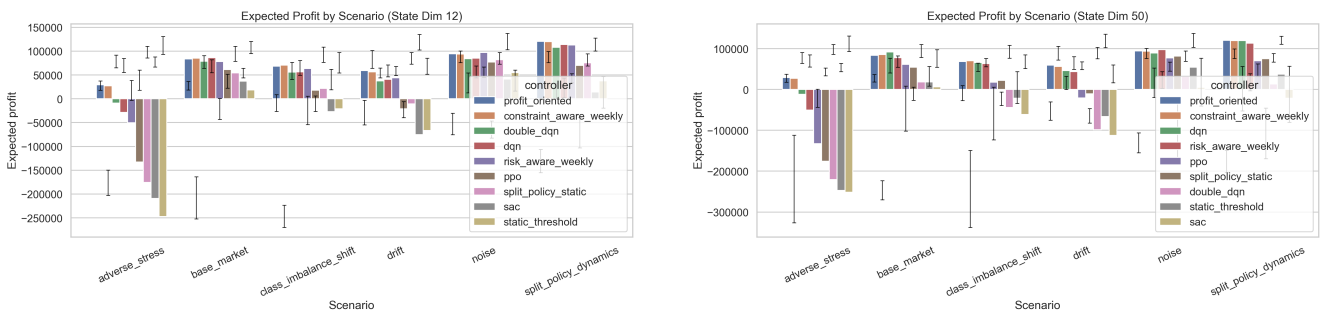


Figure 3: Default rate versus state dimension.

Figures 1–3 make the core conclusion visually clear. The overall-best baseline is flat because it does not use the varying state. By contrast, the best RL frontier rises from 12D to 30D, then flattens between 30D and 50D. The improvement is therefore real, but it is not unbounded.

5.4 Per-Dimension Scenario Plots

The repository now exports separate scenario-level figures for each state size. Figure 4 shows the expected-profit comparison for the baseline 12D state and the largest 50D state.



(a) 12D (b) 50D

Figure 4: Separate expected-profit plots exported for different observation sizes.

These per-dimension panels are useful because they preserve the scenario structure rather than collapsing everything into one cross-dimension line. They make it possible to inspect where extra state information matters and where the same baseline still dominates.

6 Discussion

6.1 Saturation

The repository-level report concludes that gains flatten by 30D to 50D. That statement is supported directly by the exported tables: the best RL expected profit rises from 59 197.47 at 12D to 66 758.29 at 30D, but then reaches only 66 946.81 at 50D. The corresponding cumulative reward moves from 67 308.36 to 78 219.69 and then stops improving. In practical terms, the 30D state captures most of the useful added signal in the quick-profile experiment.

6.2 Signal versus Complexity

The added features are not dummy variables. They encode lagged approval and default behavior, rolling profitability, segment composition, threshold motion, and realized-versus-expected gaps. The move from 12D to 30D appears to add useful information. The move from 30D to 50D mostly adds descriptive richness without a comparable increase in controller quality. This is a standard saturation pattern: beyond a certain point, more state detail adds complexity faster than it adds control value.

6.3 Stability and Threshold Behavior

In the exported quick-profile summaries, the best RL rows have essentially zero threshold-volatility mean. This implies that the selected best RL policies are either flat or nearly flat in their weekly threshold sequences despite having richer observations. That result matters for interpretation: higher-dimensional state improves controller selection quality up to 30D, but it does not automatically create more dynamically varying threshold paths.

7 Validity Checks

The dimensionality pipeline performs repository-level checks after the run. All current checks pass:

- the first 12 features remain unchanged across 12D, 20D, 30D, and 50D;
- the seeds, scenarios, controller set, reward settings, and evaluation protocol match across dimensions;
- the added features use only interactive history, last-week metrics, current week index, and fixed configuration constants;
- every controller completed the expected number of runs for every dimension;
- required CSV files and plots exist and are non-empty in each `outputs/dim_*` folder;
- the cross-dimension summary tables are internally consistent.

8 Conclusion

The controlled experiment supports a precise thesis-style conclusion. Increasing the observation size from 12 to 20 and then to 30 features improves the best RL controllers on profitability, NPV, and cumulative reward while not worsening default rate. However, the move from 30 to 50 features delivers little additional value. The evidence therefore points to a saturation effect: larger states add signal up to a point, after which they mostly add descriptive complexity rather than materially stronger control.

For the current quick-profile repository outputs, 30D is the most balanced RL state size. It captures most of the available benefit from richer portfolio history without requiring the full 50-feature observation. The strongest overall controller remains a rule-based baseline, so the main empirical message is not that reinforcement learning always wins, but that state design materially affects RL quality even when the environment, protocol, and baselines are held fixed.